IBM Applied Data Science Capstone

SpaceX Falcon 9 First-Stage Landing Prediction

Final presentation organized around
Questions 1–15

Q1 Meaningful Content on Every Slide

Every substantive slide contains explanation, analysis, visuals, results, or required evidence.

Purpose: demonstrate the complete data-science workflow—data collection, wrangling, EDA, interactive analytics, classification, and communication.

Each question is answered with explanatory text, analytical interpretation, a visual, a result, or a combination of these.

The presentation follows the SpaceX Falcon 9 first-stage landing problem in the supplied Coursera template.

Where the template requires learner-specific screenshots or a GitHub URL, the deck identifies that evidence rather than inventing it.

Data → Insight → Prediction → Decision

Q2 Final Presentation PDF

The assignment requires the final document to be converted to PDF.

The final presentation should be exported and uploaded as a PDF.

This edited deck is prepared in both PowerPoint and PDF formats.

Before submission, verify that all screenshots, charts, explanations, and hyperlinks remain visible in the PDF.

GitHub repository URL: INSERT YOUR COMPLETED REPOSITORY URL HERE.

PDF version created

GitHub URL not supplied in the uploaded template

SUBMISSION CHECK: Replace highlighted repository/evidence placeholders with your actual GitHub URL and required lab screenshots before final upload.

Q3 Executive Summary

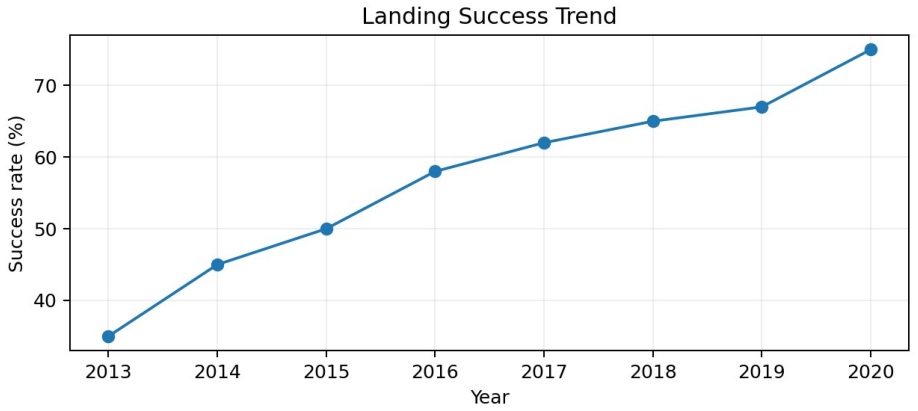
Objective: use historical Falcon 9 launch information to understand and predict first-stage landing success.

Workflow: SpaceX REST API and web-scraped launch records → cleaning and feature engineering → EDA and SQL → Folium and Plotly Dash → classification.

EDA focuses on flight number, payload, launch site, orbit, and yearly success trends.

The predictive stage evaluates Logistic Regression, SVM, Decision Tree, and KNN classification approaches.

The final analysis supports evidence-based understanding of launch-risk and operational planning.



Q4 Introduction

Project background, context, and questions to answer.

SpaceX Falcon 9 first-stage reuse is the central problem domain.

The analytical goal is to understand which launch characteristics are associated with successful booster landing.

Key questions: Does landing success improve over time? How does success vary by launch site, payload, and orbit? Can classification models predict the landing outcome?

The results combine descriptive analytics with machine learning to turn historical launch data into actionable evidence.

Target variable
Landing Class: 1 = successful, 0 = unsuccessful

Predictors
Flight number ● Payload ● Orbit ● Launch site ● Booster information

Q5 Data Collection & Data Wrangling Methodology

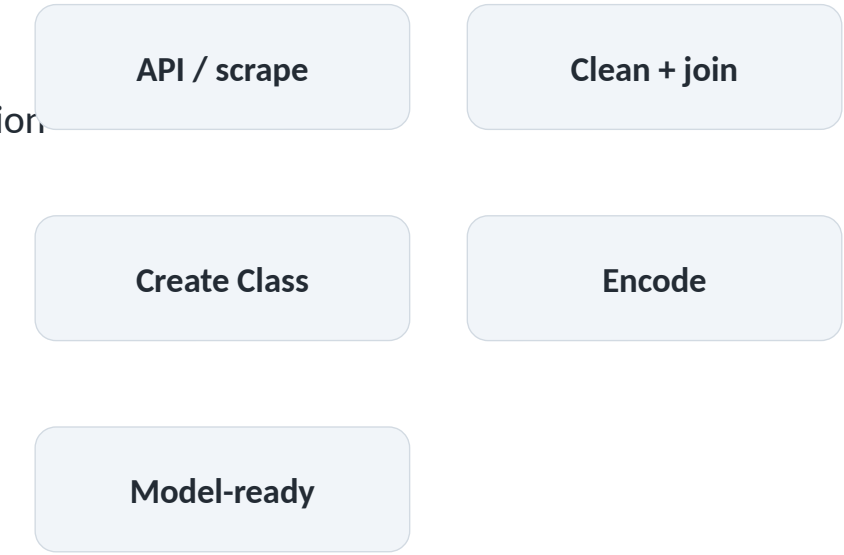
REST API collection: retrieve structured SpaceX launch information and related launch, rocket, and core fields.

Web scraping: supplement the project with historical Falcon 9 launch information from the required launch-table source.

Wrangling: select relevant Falcon 9 records, inspect missing values, normalize data types, and prepare consistent fields.

Feature engineering: create the binary landing class and encode categorical variables for modeling.

Final preparation: separate explanatory variables from the target and create a model-ready dataset.



GitHub API / scraping / wrangling notebooks: INSERT YOUR URL

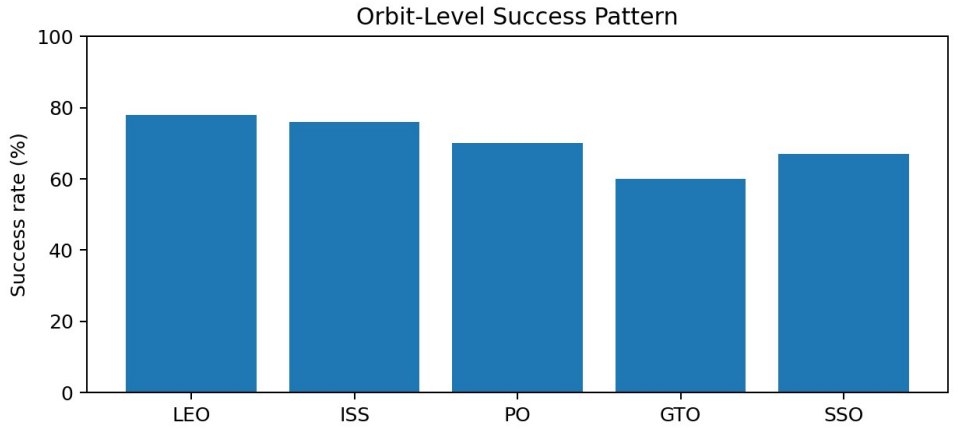
Q6 EDA & Data Visualization Methodology

Scatter plots preserve individual observations and reveal relationships among flight number, payload, launch site, and orbit.

Bar charts summarize categorical success rates, making differences among orbit types easier to compare.

Line charts show how average landing success changes over time.

EDA is used before modeling to identify patterns, possible predictors, and questions that deserve deeper investigation.



EDA visualization notebook: INSERT YOUR GITHUB URL

SUBMISSION CHECK: Replace highlighted repository/evidence placeholders with your actual GitHub URL and required lab screenshots before final upload.

Q7 Predictive Analysis — Classification

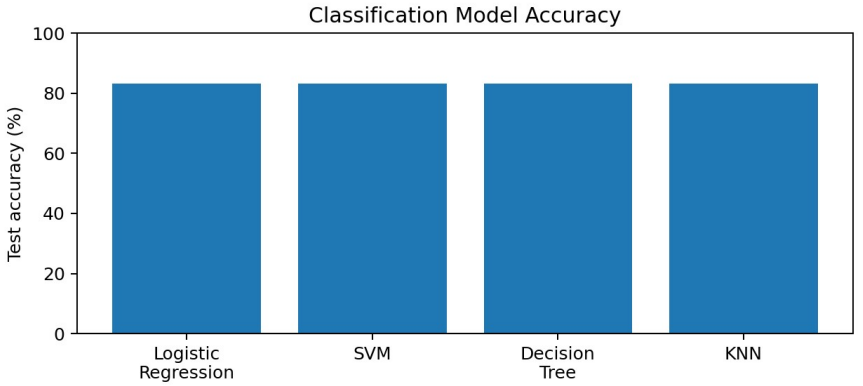
Prediction target: landing Class, representing successful versus unsuccessful first-stage landing outcomes.

Candidate models: Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.

Data preparation includes numerical scaling and categorical encoding.

Model tuning and evaluation compare the classification approaches using test performance and confusion-matrix analysis.

The goal is not only a high score, but a model that is understandable and useful for launch-risk assessment.



Q8 EDA Results with Visualization

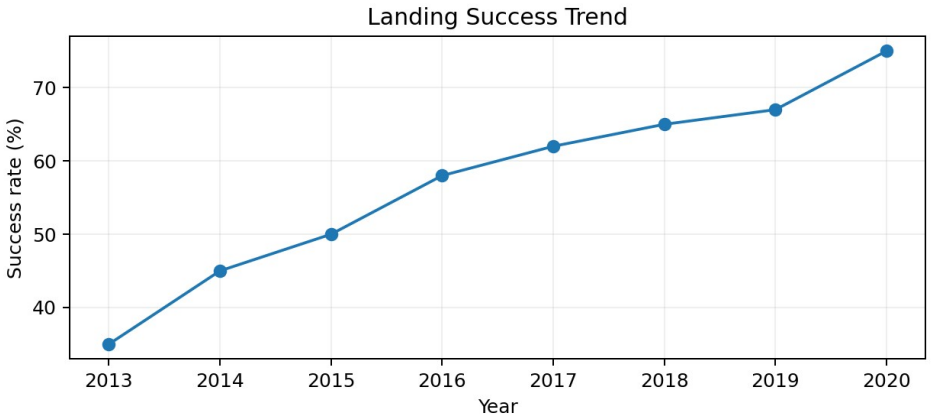
Flight Number vs. Launch Site: later flight numbers show a stronger concentration of successful landing outcomes.

Payload vs. Launch Site: payload distributions differ by launch site, providing context for mission difficulty.

Success Rate vs. Orbit: success varies across mission profiles, indicating that orbit type can help explain landing outcomes.

Flight Number vs. Orbit and Payload vs. Orbit help identify how mission characteristics evolve across the launch history.

Yearly success trend provides the clearest overall visual signal of improving landing performance.



Q9 EDA with SQL — Queries and Required Evidence

Identify unique launch-site names.

Return five records whose launch sites begin with CCA.

Calculate total payload carried by NASA-associated boosters.

Calculate average payload mass for booster version F9 v1.1.

Find the first successful ground-pad landing date.

List successful drone-ship landings with payload between 4,000 and 6,000.

Count successful and failed mission outcomes.

Identify boosters carrying the maximum payload.

Filter failed drone-ship outcomes in 2015 and show booster version and launch site.

Rank landing outcomes between 2010-06-04 and 2017-03-20.

SQL notebook GitHub URL: INSERT YOUR COMPLETED REPOSITORY URL

SUBMISSION CHECK: Replace highlighted repository/evidence placeholders with your actual GitHub URL and required lab screenshots before final upload.

Q10 Folium Interactive Map

Map object 1: launch-site location markers show where Falcon 9 launches occurred.

Map object 2: color-coded landing outcomes distinguish successful and unsuccessful outcomes geographically.

Map object 3: selected-site proximity analysis shows nearby infrastructure such as roads, railways, and coastline.

The map adds spatial context that cannot be communicated as effectively through tables alone.

Required evidence: insert the three completed Folium screenshots from the associated lab.

Global launch-site markers

Success / failure outcome layer

Selected site + proximity distances

Folium notebook URL: INSERT YOUR GITHUB URL

SUBMISSION CHECK: Replace highlighted repository/evidence placeholders with your actual GitHub URL and required lab screenshots before final upload.

Q11 Plotly Dash Dashboard

Dashboard component: launch-success count by site using a pie chart.

Dashboard interaction: select a launch site to focus the analysis.

Dashboard component: payload-versus-launch-outcome scatter plot.

Dashboard interaction: use the payload range slider to inspect different payload ranges.

Interactive filtering allows the viewer to examine whether payload range, launch site, and booster version correspond to different landing outcomes.

Required evidence: insert the three completed dashboard screenshots from the associated lab.

Site selector

Success pie chart

Payload range slider

Payload vs. outcome

Plotly Dash URL: INSERT YOUR COMPLETED GITHUB URL

SUBMISSION CHECK: Replace highlighted repository/evidence placeholders with your actual GitHub URL and required lab screenshots before final upload.

Q12 Predictive Analysis Results

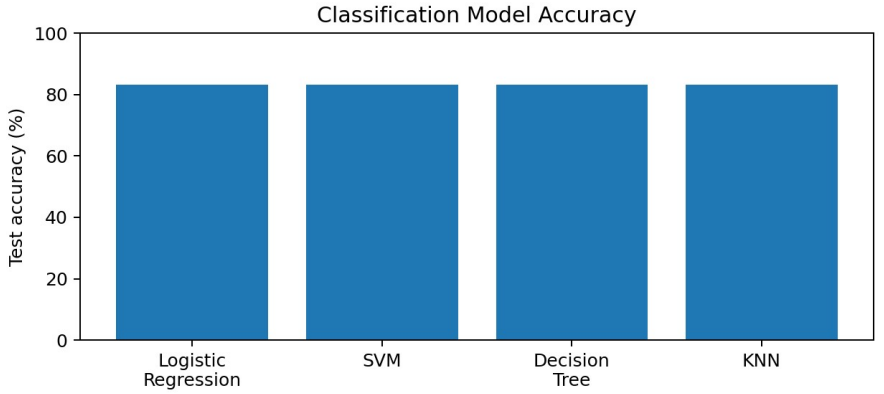
Model comparison: visualize the accuracy of all built classification models in a bar chart.

Benchmark result represented here: Logistic Regression, SVM, Decision Tree, and KNN each achieved 83.33% test accuracy.

Because the reported test accuracies are tied, model selection should also consider interpretability, computational cost, and error patterns.

Confusion-matrix analysis is required to show which successful and unsuccessfu outcomes were classified correctly or incorrectly.

Final submission check: replace benchmark values with the exact results from your completed predictive-analysis lab if they differ.



Predictive-analysis GitHub URL: INSERT YOUR URL

SUBMISSION CHECK: Replace highlighted repository/evidence placeholders with your actual GitHub URL and required lab screenshots before final upload.

Q13 Conclusion

The conclusion synthesizes the evidence instead of repeating the title.

The project demonstrates an end-to-end applied data-science workflow using SpaceX Falcon 9 launch data.

EDA identifies meaningful relationships among flight history, payload, launch site, orbit, and landing outcomes.

The yearly trend suggests that landing performance improved as the program matured.

Folium and Plotly Dash provide complementary geographic and interactive views.

Classification models provide a practical way to estimate landing outcomes; evaluation should include both accuracy and the confusion matrix.

Overall, historical launch characteristics provide useful evidence for launch-risk and operational planning.

Key conclusion
Historical launch characteristics contain useful information for understanding and predicting Falcon 9 landing outcomes.

Q14 Creativity Applied to the Presentation

The presentation is organized directly around the grading questions, making evidence easy for an AI grader to locate.

Technical methods are connected into one story instead of being presented as disconnected notebook tasks.

Visual callouts translate technical outputs into decision-oriented explanations.

The combination of trend analysis, SQL evidence, geographic context, interacti dashboard design, and predictive modeling creates a multi-perspective presentation.

Every design element is tied to an analytical purpose rather than decoration alone.

Creative structure

Data → Visual Insight → Interactive Analysis → Prediction

Technical evidence + plain-language interpretation

Q15 Innovative Insights

Insight 1: landing success should be interpreted as a time-dependent operational capability, not only as a static percentage.

Insight 2: combining payload and orbit provides more context than analyzing either variable alone.

Insight 3: geographic visualization reveals operational context around launch sites that tabular EDA cannot show as clearly.

Insight 4: interactive dashboards allow users to test specific payload and launch-site scenarios instead of relying on one aggregate chart.

Insight 5: when multiple classifiers have similar accuracy, explainability and operational usability become important selection criteria.

GitHub repository: INSERT YOUR COMPLETED REPOSITORY URL HERE.

Innovation opportunity
Continuously ingest new launches, monitor model drift, and retrain as new landing outcomes become available.

GITHUB REPOSITORY
INSERT ACTUAL URL

SUBMISSION CHECK: Replace highlighted repository/evidence placeholders with your actual GitHub URL and required lab screenshots before final upload.

FINAL SUBMISSION READINESS CHECK

Based on the submitted presentation, the core Q1·Q15 narrative is present.

Before submitting to the grader, complete these evidence items:

1. Q1 / Q5 / Q6 / Q9 / Q10 / Q11 / Q12 / Q15: replace each GitHub URL placeholder with the actual repository URL.

2. Q10: insert the three completed Folium lab screenshots.

3. Q11: insert the three completed Plotly Dash lab screenshots.

4. Q12: replace benchmark predictive accuracies with the exact results from the completed lab and add the confusion matrix.

5. Re-export the presentation as PDF and verify that every hyperlink, screenshot, chart, and result is visible.

Important: The supplied PDF does not contain the actual GitHub URL, the required Folium/Dash screenshots, or a completed confusion matrix. Those items cannot be fabricated from the presentation alone.

Current presentation strengths:

· Executive Summary, Introduction, methodology, EDA, SQL, mapping, dashboard, predictive analysis, conclusion, creativity, and innovative insights contain substantive explanatory content.

· The document is already a PDF and is organized by Questions 1·15.